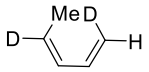
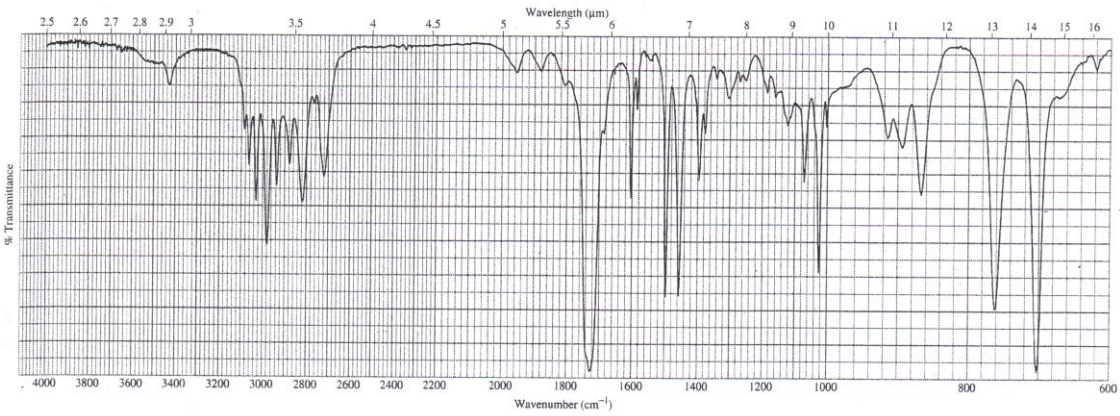
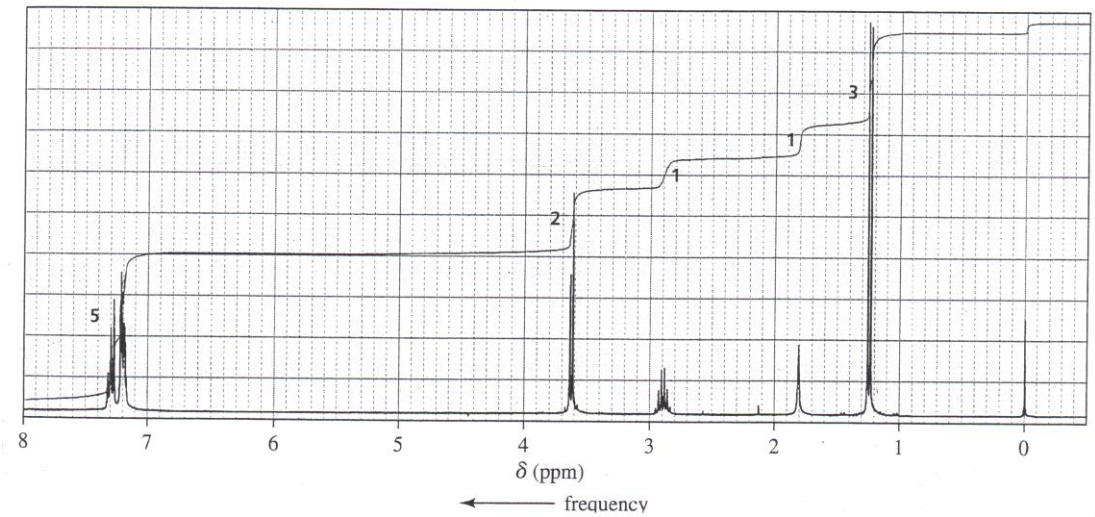
Organic Cumulative Examination

March 8, 2024

1. Give at least two examples for the use of ^{195}Pt NMR spectroscopy (10 points each).

2. Explain at least two different methods to separate a racemate of D- and L-glucosamine into the pure enantiomers in a preparative scale (industrially relevant), and explain the underlying principles (15 points each)

3. The following compound is heated at 127 °C for 49 days. Only one compound, an open chain isomer, is isolated afterwards. Which one? What is the mechanism for the transformation (20 points) ?



4. A compound gives the following IR spectrum. Upon reaction with sodium borohydride followed by acidification, it forms the product with the following ^{1}H NMR spectrum. Identify all compounds (30 points)